POWER PULSE

Surge Impedance Loading (SIL) Important Formulae

Notes by Balaji Sir

1 Surge impedance and SIL

1. $$Z_c = \sqrt{\frac{L}{C}} = \sqrt{\frac{z}{y}} \text{ (lossless)}$$

   ≈ 250–400 Ω for overhead lines

2. $$SIL = \frac{V_{LL}^2}{Z_c} \text{ MW} \quad (V_{LL} \text{ in kV},\ Z_c \text{ in } \Omega)$$

3. $$\text{At SIL: } Q_{gen} = \omega C V^2 = Q_{abs} = \omega L I^2 \Rightarrow \text{ flat voltage profile}$$

4. $$P < SIL: \text{ line generates vars, } V_r \uparrow; \qquad P > SIL: \text{ line absorbs vars, } V_r \downarrow$$

2 Wave quantities

5. $$\beta = \omega\sqrt{LC}, \quad v = \frac{1}{\sqrt{LC}}, \quad \lambda = \frac{v}{f}, \quad \theta = \beta l$$

   electrical length

6. $$P_{max} = \frac{V_s V_r}{Z_c \sin \beta l} = \frac{SIL}{\sin \beta l} \quad (V_s = V_r)$$

   lossless line

3 Effect of compensation

7. $$\text{Series } k_{se}: \quad Z_c' = Z_c\sqrt{1 - k_{se}}, \quad SIL' = \frac{SIL}{\sqrt{1 - k_{se}}}$$

   SIL increases

8. $$\text{Shunt } k_{sh}: \quad Z_c' = \frac{Z_c}{\sqrt{1 - k_{sh}}}, \quad SIL' = SIL\sqrt{1 - k_{sh}}$$

   SIL decreases

9. $$\text{Bundling: } L \downarrow,\ C \uparrow \Rightarrow\ Z_c \downarrow,\ SIL \uparrow$$

10. $\text{SIL} \propto V^2$ (approx. 130 MW at 220 kV, 500 MW at 400 kV, 2300 MW at 765 kV)